DevOps & Networking Fundamentals

1. Introduction to DevOps

1.1 Understanding the SDLC

SDLC (Software Development Life Cycle) is the structured process used to develop software efficiently.

Stages:

• Planning – Identify requirements and project goals

• Design – Architecture and system design

• Development – Writing the code

• Testing – Checking for bugs and errors

• Deployment – Releasing the application

• Maintenance – Updates and bug fixes

DevOps improves SDLC by automating development and operations workflows.

1.2 What is Agile Methodology?

Agile is a modern software development methodology that focuses on:

• Iterative development

• Collaboration

• Continuous feedback

• Faster delivery

Instead of long development cycles, Agile uses short cycles called sprints.

Popular Agile frameworks:

• Scrum

• Kanban

Benefits:

• Faster product delivery

• Better team collaboration

• Easier adaptation to changes

1.3 DevOps Decoded

DevOps = Development + Operations

It is a culture and set of practices that combine software development and IT operations to deliver applications faster and more reliably.

Key goals:

• Continuous Integration (CI)

• Continuous Delivery (CD)

• Automation

• Monitoring

• Collaboration

Popular DevOps tools:

• Git / GitHub

• Jenkins

• Docker

• Kubernetes

• AWS

1.4 DevOps Process Explained

Typical DevOps lifecycle:

1. Plan
2. Develop
3. Build
4. Test
5. Release
6. Deploy
7. Operate
8. Monitor

This cycle enables continuous improvement and faster software delivery.

Networking Fundamentals for DevOps

2.1 What is Networking?

Networking is the process of connecting computers and devices to share data and resources.

Examples:

• Internet

• Local Area Network (LAN)

• Cloud networks

Networking enables communication between:

• Servers

• Applications

• Users

2.2 OSI Model Demystified

The OSI (Open Systems Interconnection) model is a conceptual framework used to understand network communication.

It has 7 layers:

1. Application

2. Presentation

3. Session

4. Transport

5. Network

6. Data Link

7. Physical

2.3 OSI vs TCP/IP Model

OSI Model – 7 layers, conceptual model.

TCP/IP Model – 4 layers, practical internet model.

TCP/IP Layers:

1. Application
2. Transport
3. Internet
4. Network Access

2.4 What is IP?

IP (Internet Protocol) identifies devices on a network.

Example IP address:

192.168.1.1

Types:

- IPv4 – 32-bit address
- IPv6 – 128-bit address

2.5 DNS Explained

DNS (Domain Name System) converts domain names into IP addresses.

Example:

google.com → 142.250.190.78

Without DNS, users would need to remember IP addresses.

2.6 Subnetting Simplified

Subnetting divides a network into smaller networks called subnets.

Purpose:

• Improve network performance

• Increase security

• Efficient IP management

Example network:

192.168.1.0/24

2.7 Port Numbers Simplified

Ports allow multiple services to run on a single server.

Common ports:

22 – SSH

80 – HTTP

443 – HTTPS

21 – FTP

25 – SMTP

2.8 HTTP Uncovered

HTTP (Hypertext Transfer Protocol) allows communication between web browsers and web servers.

Browser → HTTP Request → Server

Server → HTTP Response → Browser

Secure version:

HTTPS (HTTP Secure) using SSL/TLS encryption.